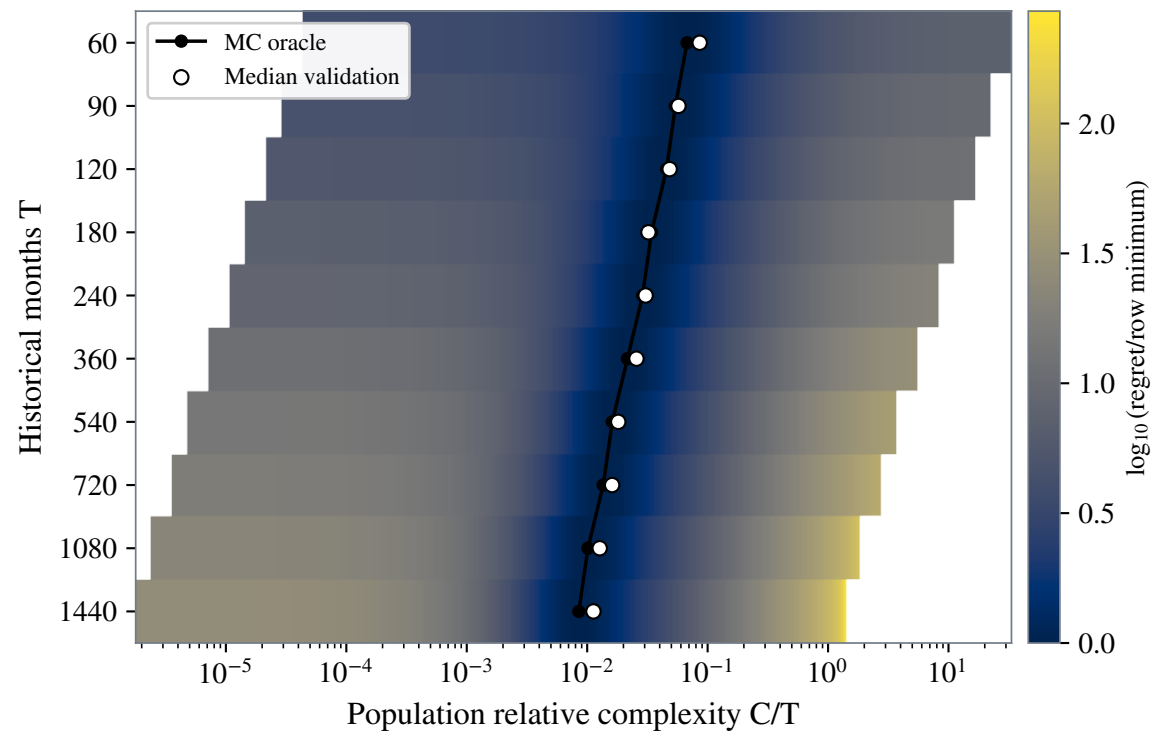


Appendix: smooth nonlinear economy

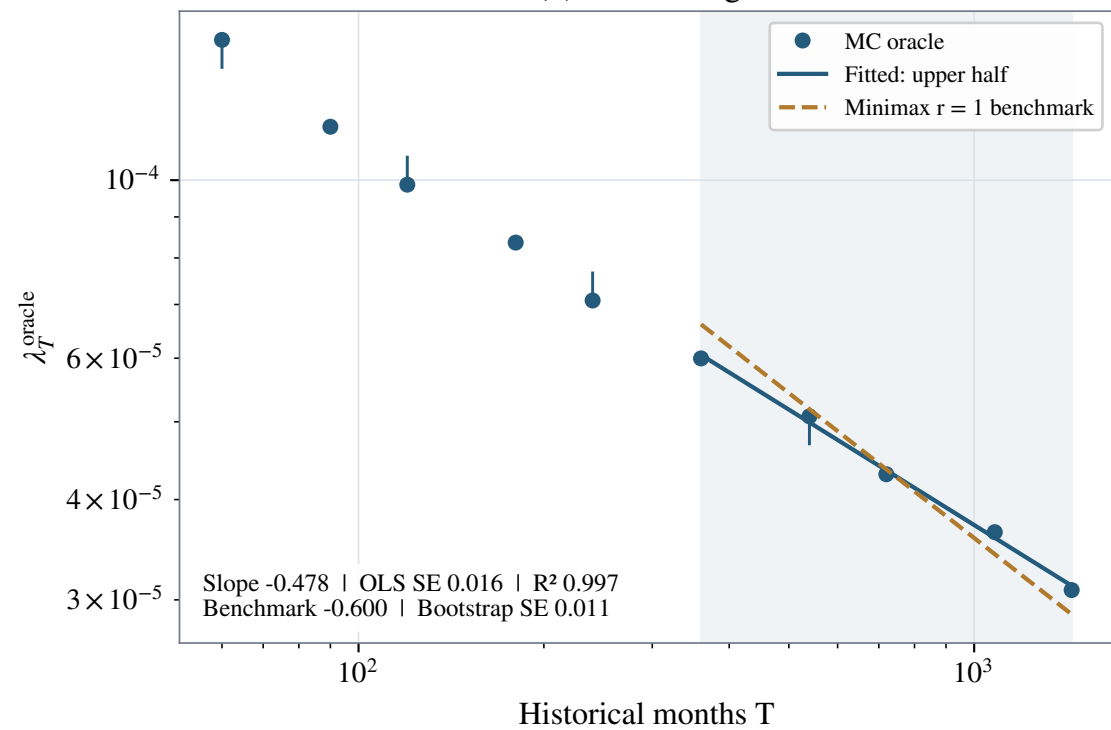
$b = 1.5$ | 5,000 stocks | 2,000 directions | 500 independent paths

Target: $1/j$. This smooth nonlinear economy need not attain the worst-case $r = 1$ rate.

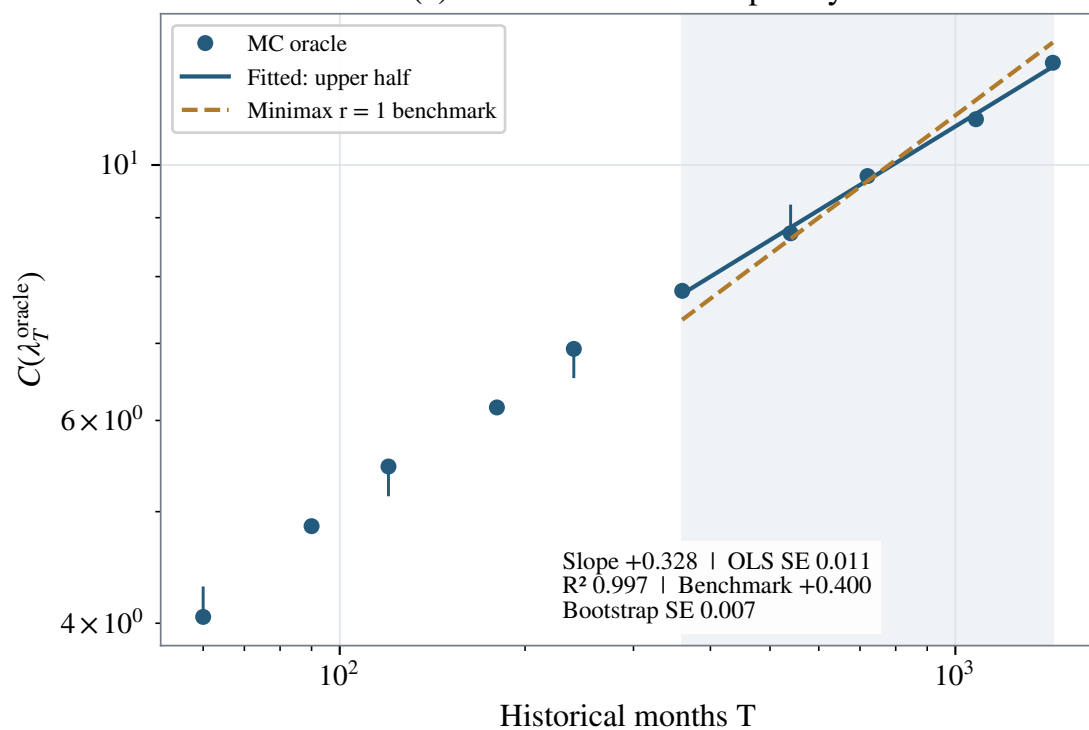
(a) Finite-sample learnability map



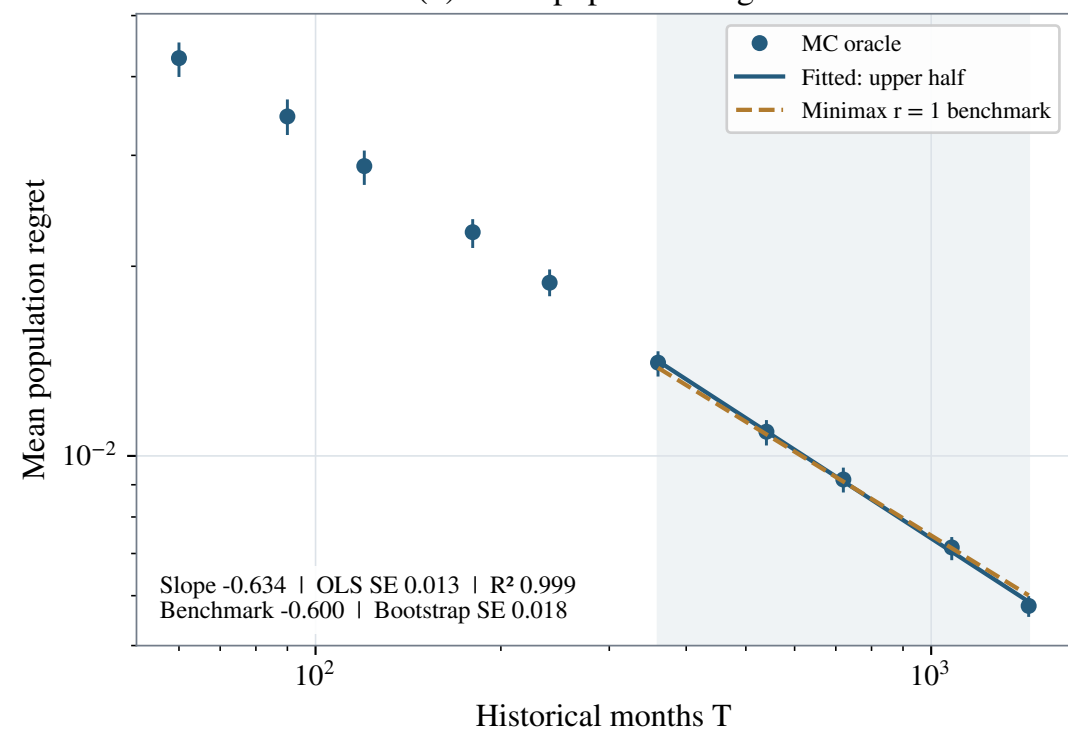
(b) Oracle ridge



(c) Oracle effective complexity



(d) Exact population regret



Risk is evaluated analytically, not on a simulated test set. Bars: 95% bootstrap intervals across independent paths.

Slope fits use the predeclared upper half, $T = 360$ -1,440 (shaded); all sample sizes remain visible.

Dashed slopes are minimax $r = 1$ benchmarks, not predicted DGP-specific oracle rates.

The full C/T range is shown on a log axis and can exceed one. Median validation markers do not describe mean validation risk.